

Exercise Set 1.1

- Determine whether each equation is linear in x_1, x_2, x_3 .
(a) $x_1 + 5x_2 - \sqrt{2}x_3 = 1$ (b) $x_1 + 3x_2 + x_1x_3 = 2$ (c) $x_1 = -7x_2 + 3x_3$
(d) $x_1^{-2} + x_2 + 8x_3 = 5$ (e) $x_1^{3/5} - 2x_2 + x_3 = 4$ (f) $\pi x_1 - \sqrt{2}x_2 = 7^{1/3}$
- Using the notation of Formula (7), write a general linear system with
(a) two equations in two unknowns. (b) three equations in three unknowns. (c) two equations in four unknowns.
- Find a system of linear equations in $x_1, x_2, x_3, \dots$ corresponding to each augmented matrix.
(a) $\begin{bmatrix} 2 & 0 & 0 \\ 3 & -4 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 0 & -2 & 5 \\ 7 & 1 & 4 & -3 \\ 0 & -2 & 1 & 7 \end{bmatrix}$
- Find the augmented matrix for each linear system.
(a) $-2x_1 = 6, 3x_1 = 8, 9x_1 = -3$ (b) $6x_1 - x_2 + 3x_3 = 4, 5x_2 - x_3 = 1$
(c) $2x_2 - 3x_4 + x_5 = 0, -3x_1 - x_2 + x_3 = -1, 6x_1 + 2x_2 - x_3 + 2x_4 - 3x_5 = 6$
- Determine whether each 3-tuple is a solution of

$$2x_1 - 4x_2 - x_3 = 1, \quad x_1 - 3x_2 + x_3 = 1, \quad 3x_1 - 5x_2 - 3x_3 = 1$$

- (a) $(3, 1, 1)$ (b) $(3, -1, 1)$ (c) $(13, 5, 2)$ (d) $(\frac{13}{2}, \frac{5}{2}, 2)$ (e) $(17, 7, 5)$
- Solve each system, if possible, and state whether the corresponding lines have zero, one, or infinitely many intersection points; give coordinates or parametric equations accordingly.
(a) $3x - 2y = 4, 6x - 4y = 9$ (b) $2x - 4y = 1, 4x - 8y = 2$ (c) $x - 2y = 0, x - 4y = 8$
- Use parametric equations to describe the solution set of each linear equation.
(a) $7x - 5y = 3$ (b) $3x_1 - 5x_2 + 4x_3 = 7$ (c) $-8x_1 + 2x_2 - 5x_3 + 6x_4 = 1$ (d) $3v - 8w + 2x - y + 4z = 0$
- Each system below has infinitely many solutions; describe each solution set parametrically.
(a) $2x - 3y = 1, 6x - 9y = 3$ (b) $x_1 + 3x_2 - x_3 = -4, 3x_1 + 9x_2 - 3x_3 = -12, -x_1 - 3x_2 + x_3 = 4$
- Find one elementary row operation that creates a 1 in the upper-left corner of the given augmented matrix without introducing fractions in the first row.
(a) $\begin{bmatrix} -3 & -1 & 2 & 4 \\ 2 & -3 & 3 & 2 \\ 0 & 2 & -3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & -1 & -5 & 0 \\ 2 & -9 & 3 & 2 \\ 1 & 4 & -3 & 3 \end{bmatrix}$
- Find all values of k for which the given augmented matrix corresponds to a consistent linear system.
(a) $\begin{bmatrix} 1 & k & -4 \\ 4 & 8 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & k & -1 \\ 4 & 8 & -4 \end{bmatrix}$
- The curve $y = ax^2 + bx + c$ shown passes through $(x_1, y_1), (x_2, y_2), (x_3, y_3)$. Show that a, b, c solve the system with augmented matrix

$$\begin{bmatrix} x_1^2 & x_1 & 1 & y_1 \\ x_2^2 & x_2 & 1 & y_2 \\ x_3^2 & x_3 & 1 & y_3 \end{bmatrix}$$

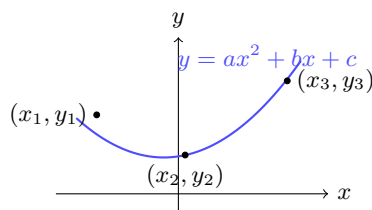


FIGURE Ex-21

22. Explain why none of the three elementary row operations changes the solution set of a linear system.
23. Show that if $x_1 + kx_2 = c$ and $x_1 + lx_2 = d$ have the same solution set, then $k = l$ and $c = d$.
25. A diet requires 7 units of fat, 9 of protein, and 16 of carbohydrates per meal, drawn from three foods (per ounce):
- Food 1: 2 fat, 2 protein, 4 carbohydrates.
- Food 2: 3 fat, 1 protein, 2 carbohydrates.
- Food 3: 1 fat, 3 protein, 5 carbohydrates.
- With x, y, z the ounces of foods 1, 2, 3, find (but do not solve) the linear system whose solution gives the required amounts.

Exercise Set 1.2

1. Classify each matrix as row echelon form, reduced row echelon form, both, or neither.

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & 4 \end{bmatrix}$ (e) $\begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

(f) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ (g) $\begin{bmatrix} 1 & -7 & 5 & 5 \\ 0 & 1 & 3 & 2 \end{bmatrix}$

3. Each augmented matrix below is a row echelon form reached by row operations. Identify its pivot rows/columns and solve the system.

(a) $\begin{bmatrix} 1 & -3 & 4 & 7 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 7 & -2 & 0 & -8 & -3 \\ 0 & 0 & 1 & 1 & 6 & 5 \\ 0 & 0 & 0 & 1 & 3 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & -3 & 7 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

5. Solve by Gaussian elimination.

$$x_1 + x_2 + 2x_3 = 8, \quad -x_1 - 2x_2 + 3x_3 = 1, \quad 3x_1 - 7x_2 + 4x_3 = 10$$

7. Solve by Gaussian elimination.

$$x - y + 2z - w = -1, \quad 2x + y - 2z - 2w = -2, \quad -x + 2y - 4z + w = 1, \quad 3x - 3w = -3$$

9. Solve Exercise 5 by Gauss–Jordan elimination.

11. Solve Exercise 7 by Gauss–Jordan elimination.

13. Determine by inspection (no pencil and paper) whether the homogeneous system has nontrivial solutions.

$$2x_1 - 3x_2 + 4x_3 - x_4 = 0, \quad 7x_1 + x_2 - 8x_3 + 9x_4 = 0, \quad 2x_1 + 8x_2 + x_3 - x_4 = 0$$

15. Solve by any method.

$$2x_1 + x_2 + 3x_3 = 0, \quad x_1 + 2x_2 = 0, \quad x_2 + x_3 = 0$$

17. Solve by any method.

$$3x_1 + x_2 + x_3 + x_4 = 0, \quad 5x_1 - x_2 + x_3 - x_4 = 0$$

21. Solve by any method.

$$2I_1 - I_2 + 3I_3 + 4I_4 = 9, \quad I_1 - 2I_3 + 7I_4 = 11, \quad 3I_1 - 3I_2 + I_3 + 5I_4 = 8, \quad 2I_1 + I_2 + 4I_3 + 4I_4 = 10$$

23. Each augmented matrix below has unspecified entries *. Determine whether the system is consistent and, if so, whether the solution is unique – answer “inconclusive” if there isn’t enough information.

$$(a) \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{bmatrix} \quad (b) \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & * & * & * \\ 0 & 0 & * & 0 \\ 0 & 0 & 1 & * \end{bmatrix}$$

25. Determine the values of a for which the system has no solution, exactly one solution, or infinitely many solutions.

$$x + 2y - 3z = 4, \quad 3x - y + 5z = 2, \quad 4x + y + (a^2 - 14)z = a + 2$$

27. What condition, if any, must a, b, c satisfy for the system to be consistent?

$$x + 3y - z = a, \quad x + y + 2z = b, \quad 2y - 3z = c$$

31. Find two different row echelon forms of

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

– showing that row echelon forms need not be unique.

32. Reduce

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 3 & 4 & 5 \end{bmatrix}$$

to reduced row echelon form without introducing fractions at any intermediate stage.

33. Show that the following nonlinear system has 18 solutions for $0 \leq \alpha, \beta, \gamma \leq 2\pi$.

$$\sin \alpha + 2 \cos \beta + 3 \tan \gamma = 0, \quad 2 \sin \alpha + 5 \cos \beta + 3 \tan \gamma = 0, \quad -\sin \alpha - 5 \cos \beta + 5 \tan \gamma = 0$$

[Hint: substitute $x = \sin \alpha$, $y = \cos \beta$, $z = \tan \gamma$.]

35. Solve the following nonlinear system for x, y, z .

$$x^2 + y^2 + z^2 = 6, \quad x^2 - y^2 + 2z^2 = 2, \quad 2x^2 + y^2 - z^2 = 3$$

[Hint: substitute $X = x^2$, $Y = y^2$, $Z = z^2$.]

37. Find a, b, c, d so that the curve shown is the graph of $y = ax^3 + bx^2 + cx + d$.

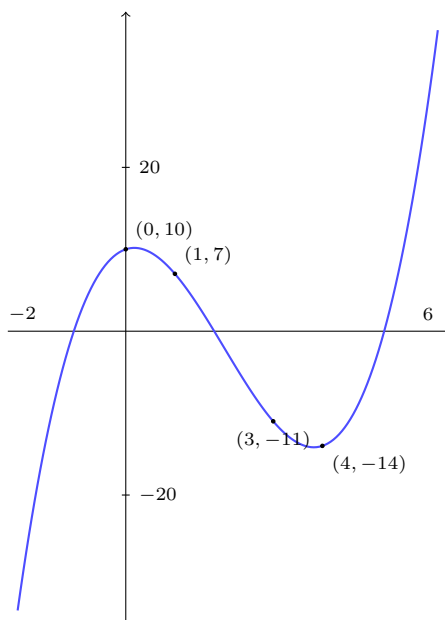


FIGURE Ex-37

39. If the linear system

$$a_1x + b_1y + c_1z = 0, \quad a_2x - b_2y + c_2z = 0, \quad a_3x + b_3y - c_3z = 0$$

has only the trivial solution, what can be said about the solutions of

$$a_1x + b_1y + c_1z = 3, \quad a_2x - b_2y + c_2z = 7, \quad a_3x + b_3y - c_3z = 11 ?$$

40. (a) If A has three rows and five columns, what is the maximum possible number of leading 1's in its reduced row echelon form?
 (b) If B has three rows and six columns, what is the maximum possible number of parameters in the general solution of the system with augmented matrix B ?
 (c) If C has five rows and three columns, what is the minimum possible number of zero rows in any row echelon form of C ?

41. Describe all possible reduced row echelon forms of

$$(a) \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad (b) \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & p & q \end{bmatrix}$$

Exercise Set 1.3

1. Suppose A, B, C, D, E have sizes $(4 \times 5), (4 \times 5), (5 \times 2), (4 \times 2), (5 \times 4)$. Determine whether each expression is defined, and if so, give the size of the result.

(a) BA (b) AB^T (c) $AC + D$ (d) $E(AC)$ (e) $A - 3E^T$ (f) $E(5B + A)$

3. Using $A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$, $E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$,

compute if defined:

(a) $D + E$ (b) $D - E$ (c) $5A$ (d) $-7C$ (e) $2B - C$ (f) $4E - 2D$
 (g) $-3(D + 2E)$ (h) $A - A$ (i) $\text{tr}(D)$ (j) $\text{tr}(D - 3E)$ (k) $4 \text{tr}(7B)$ (l) $\text{tr}(A)$

4. Using the matrices from Exercise 3, compute if defined:

(a) $2A^T + C$ (b) $D^T - E^T$ (c) $(D - E)^T$ (d) $B^T + 5C^T$ (e) $\frac{1}{2}C^T - \frac{1}{4}A$ (f) $B - B^T$
 (g) $2E^T - 3D^T$ (h) $(2E^T - 3D^T)^T$ (i) $(CD)E$ (j) $C(BA)$ (k) $\text{tr}(DE^T)$ (l) $\text{tr}(BC)$

7. Using $A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$, use the row or column method to find:

(a) the first row of AB (b) the third row of AB (c) the second column of AB
 (d) the first column of BA (e) the third row of AA (f) the third column of AA

9. Using A, B from Exercise 7:

(a) Express each column of AA as a linear combination of the columns of A .
 (b) Express each column of BB as a linear combination of the columns of B .

11. Find matrices $A, \mathbf{x}, \mathbf{b}$ expressing each system as $A\mathbf{x} = \mathbf{b}$, and write out the equation.

(a) $2x_1 - 3x_2 + 5x_3 = 7$, $9x_1 - x_2 + x_3 = -1$, $x_1 + 5x_2 + 4x_3 = 0$
 (b) $4x_1 - 3x_3 + x_4 = 1$, $5x_1 + x_2 - 8x_4 = 3$, $2x_1 - 5x_2 + 9x_3 - x_4 = 0$, $3x_2 - x_3 + 7x_4 = 2$

13. Express each matrix equation as a system of linear equations.

(a) $\begin{bmatrix} 5 & 6 & -7 \\ -1 & -2 & 3 \\ 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 5 & -3 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -9 \end{bmatrix}$

15. Find all values of k , if any, satisfying

$$\begin{bmatrix} k & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

17. Use the column-row expansion of AB to express the product as a sum of matrix products, for

$$A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix}.$$

21. For the linear system in Example 5 of Section 1.2, express the general solution obtained there as a linear combination of column vectors containing only numerical entries. [Suggestion: write the general solution as one column vector, split it into column vectors each containing at most one parameter, then factor out the parameters.]

23. Solve for a, b, c, d :

$$\begin{bmatrix} a & 3 \\ -1 & a+b \end{bmatrix} = \begin{bmatrix} 4 & d-2c \\ d+2c & -2 \end{bmatrix}$$

25. (a) Show that if A has a row of zeros and B is any matrix for which AB is defined, then AB also has a row of zeros.

(b) State and justify a similar result for a column of zeros.

27. How many 3×3 matrices A satisfy $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ x-y \\ 0 \end{bmatrix}$ for all x, y, z ?

29. A matrix B is a **square root** of A if $BB = A$.

(a) Find two square roots of $A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$.

(b) How many different square roots can you find of $A = \begin{bmatrix} 5 & 0 \\ 0 & 9 \end{bmatrix}$?

(c) Do you think every 2×2 matrix has at least one square root? Explain.

30. Let 0 denote the 2×2 zero matrix.

(a) Is there a 2×2 matrix $A \neq 0$ with $AA = 0$? Justify your answer.

(b) Is there a 2×2 matrix $A \neq 0$ with $AA = A$? Justify your answer.

32. Find a 4×4 matrix $A = [a_{ij}]$ satisfying:

$$(a) a_{ij} = i + j \quad (b) a_{ij} = i^{j-1} \quad (c) a_{ij} = \begin{cases} 1 & |i-j| > 1 \\ -1 & |i-j| \leq 1 \end{cases}$$

33. Type I, II, III items cost \$1, \$2, \$3 each; the table gives the numbers purchased each month:

	Type I	Type II	Type III
Jan.	3	4	3
Feb.	5	6	0
Mar.	2	9	4
Apr.	1	1	7

What does the following product represent?

$$\begin{bmatrix} 3 & 4 & 3 \\ 5 & 6 & 0 \\ 2 & 9 & 4 \\ 1 & 1 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

35. **(Working with Proofs)** Prove: if A, B are $n \times n$, then $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$.

Exercise Set 1.4

1. For $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ 1 & -4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$, $a = 4, b = -7$, verify:

(a) the associative law for addition (b) the associative law for multiplication

(c) the left distributive law (d) $(a+b)C = aC + bC$

3. Using the matrices from Exercise 1, verify:
 (a) $(A^T)^T = A$ (b) $(AB)^T = B^T A^T$
5. Use Theorem 1.4.5 to find the inverse of $A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$.
10. Find the inverse of $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$.
13. For $A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, verify $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.
15. Given $(7A)^{-1} = \begin{bmatrix} -3 & 7 \\ 1 & -2 \end{bmatrix}$, find A .
19. For $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, compute (a) A^3 (b) A^{-3} (c) $A^2 - 2A + I$.
21. For $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, compute $p(A)$ for
 (a) $p(x) = x - 2$ (b) $p(x) = 2x^2 - x + 1$ (c) $p(x) = x^3 - 2x + 1$
23. For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, find all a, b, c, d for which A, B commute.
25. Use the method of Example 8 to find the unique solution of
 $3x_1 - 2x_2 = -1$, $4x_1 + 5x_2 = 3$
30. Verify $p(A) = p_1(A)p_2(A)$ for $p(x) = x^2 - 9 = (x + 3)(x - 3)$ and an arbitrary square matrix A .
31. (a) Give an example of 2×2 matrices with $(A + B)(A - B) \neq A^2 - B^2$.
 (b) State a valid formula for $(A + B)(A - B)$.
 (c) What condition on A, B allows $(A + B)(A - B) = A^2 - B^2$?
32. The equation $a^2 = 1$ has exactly two real solutions. Find at least eight solutions of $A^2 = I_3$. [*Hint*: look for solutions with all off-diagonal entries zero.]
33. (a) Show that if $A^2 + 2A + I = 0$, then A is invertible; find A^{-1} .
 (b) Show that if $p(x)$ has nonzero constant term and $p(A) = 0$, then A is invertible.
34. Is it possible for A^3 to equal the identity matrix without A being invertible? Explain.
35. Can a matrix with a zero row or column have an inverse? Explain.
36. Can a matrix with two identical rows or columns have an inverse? Explain.
37. Determine whether $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ is invertible, and if so find A^{-1} . [*Hint*: solve $AX = I$ for X by equating corresponding entries.]
39. Simplify, assuming A, B, C, D invertible: $(AB)^{-1}(AC^{-1})(D^{-1}C^{-1})^{-1}D^{-1}$.
41. Show that if R is $1 \times n$ and C is $n \times 1$, then $RC = \text{tr}(CR)$.

43. (a) Show that if A is invertible and $AB = AC$, then $B = C$.
 (b) Explain why part (a) and Example 3 do not contradict each other.
45. (a) Show that if $A, B, A + B$ are invertible matrices of the same size, then $A(A^{-1} + B^{-1})B(A + B)^{-1} = I$.
 (b) What does part (a) tell you about the matrix $A^{-1} + B^{-1}$?
46. A square matrix A is **idempotent** if $A^2 = A$.
 (a) Show that if A is idempotent, so is $I - A$.
 (b) Show that if A is idempotent, then $2A - I$ is invertible and is its own inverse.
47. Show that if $A^k = 0$ for some positive integer k , then $I - A$ is invertible and $(I - A)^{-1} = I + A + A^2 + \cdots + A^{k-1}$.
48. Show that $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfies $A^2 - (a + d)A + (ad - bc)I = 0$.
51. **(Working with Proofs)** Prove Theorem 1.4.1(a).

Exercise Set 1.5

1. Determine whether the matrix is elementary.

(a) $\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -5 & 1 \\ 1 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

3. Find a row operation and its elementary matrix that restores the given elementary matrix to I .

(a) $\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

5. Identify the row operation corresponding to E and verify that EA results from applying it to A .

(a) $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

(b) $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}, A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$

(c) $E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

7. Using $A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}, B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$, find an elementary matrix E satisfying:

(a) $EA = B$ (b) $EB = A$ (c) $EA = C$ (d) $EC = A$

9. First use Theorem 1.4.5, then the inversion algorithm, to find A^{-1} if it exists.

$$(a) A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix} \quad (b) A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

11. Use the inversion algorithm to find the inverse, if it exists.

$$(a) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \quad (b) \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

13. Use the inversion algorithm to find the inverse, if it exists, of $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

19. Find the inverse of each 4×4 matrix, where k_1, k_2, k_3, k_4, k are all nonzero.

$$(a) \begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix} \quad (b) \begin{bmatrix} k & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & k & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

21. Find all values of c , if any, for which the matrix is invertible.

$$\begin{bmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{bmatrix}$$

23. Express the matrix and its inverse as products of elementary matrices.

$$\begin{bmatrix} -3 & 1 \\ 2 & 2 \end{bmatrix}$$

27. Show that $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 1 \\ 2 & 1 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 2 & -2 \\ 1 & 1 & 4 \end{bmatrix}$ are row equivalent by finding a sequence of elementary row operations that produces B from A , and use it to find a matrix C with $CA = B$.

29. Show that if $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{bmatrix}$ is an elementary matrix, then at least one entry in the third row must be zero.

30. Show that

$$A = \begin{bmatrix} 0 & a & 0 & 0 & 0 \\ b & 0 & c & 0 & 0 \\ 0 & d & 0 & e & 0 \\ 0 & 0 & f & 0 & g \\ 0 & 0 & 0 & h & 0 \end{bmatrix}$$

is not invertible for any values of the entries.

31. **(Working with Proofs)** Prove that if A, B are $m \times n$ matrices, then A, B are row equivalent if and only if they have the same reduced row echelon form.

32. **(Working with Proofs)** Prove that if A is invertible and B is row equivalent to A , then B is also invertible.
33. **(Working with Proofs)** Prove that if B is obtained from A by a sequence of elementary row operations, then a second sequence of elementary row operations, applied to B , recovers A .

Exercise Set 1.6

1. Solve by inverting the coefficient matrix and using Theorem 1.6.2.
 $x_1 + x_2 = 2, \quad 5x_1 + 6x_2 = 9$
3. $x_1 + 3x_2 + x_3 = 4, \quad 2x_1 + 2x_2 + x_3 = -1, \quad 2x_1 + 3x_2 + x_3 = 3$
6. $-x - 2y - 3z = 0, \quad w + x + 4y + 4z = 7, \quad w + 3x + 7y + 9z = 4, \quad -w - 2x - 4y - 6z = 6$
7. Solve for x_1, x_2 in terms of b_1, b_2 : $3x_1 + 5x_2 = b_1, \quad x_1 + 2x_2 = b_2$
9. Solve the linear systems together by reducing an appropriate augmented matrix to reduced row echelon form.
 $x_1 - 5x_2 = b_1, \quad 3x_1 + 2x_2 = b_2$
 (i) $b_1 = 1, b_2 = 4$ (ii) $b_1 = -2, b_2 = 5$
13. Determine conditions on the b_i 's, if any, for the system to be consistent.
 $x_1 + 3x_2 = b_1, \quad -2x_1 + x_2 = b_2$
15. $x_1 - 2x_2 + 5x_3 = b_1, \quad 4x_1 - 5x_2 + 8x_3 = b_2, \quad -3x_1 + 3x_2 - 3x_3 = b_3$
18. For $A = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & -2 \\ 3 & 1 & 1 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$:
 (a) Show that $A\mathbf{x} = \mathbf{x}$ can be rewritten as $(A - I)\mathbf{x} = \mathbf{0}$, and use this to solve $A\mathbf{x} = \mathbf{x}$.
 (b) Solve $A\mathbf{x} = 4\mathbf{x}$.
19. Solve the matrix equation for X .

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 0 \\ 0 & 2 & -1 \end{bmatrix} X = \begin{bmatrix} 2 & -1 & 5 & 7 & 8 \\ 4 & 0 & -3 & 0 & 1 \\ 3 & 5 & -7 & 2 & 1 \end{bmatrix}$$
21. **(Working with Proofs)** Let $A\mathbf{x} = \mathbf{0}$ be a homogeneous system of n equations in n unknowns with only the trivial solution. Prove that for any positive integer k , $A^k\mathbf{x} = \mathbf{0}$ also has only the trivial solution.
23. **(Working with Proofs)** Let $A\mathbf{x} = \mathbf{b}$ be any consistent system, and let $\mathbf{x}_1$ be a fixed solution. Prove that every solution can be written $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_0$ where $\mathbf{x}_0$ solves $A\mathbf{x} = \mathbf{0}$, and that every matrix of this form is a solution.
24. **(Working with Proofs)** Use part (a) of Theorem 1.6.3 to prove part (b).

Exercise Set 1.7

1. Classify each matrix as upper triangular, lower triangular, diagonal (note: a diagonal matrix is both), or none of these.

$$(a) \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \quad (b) \begin{bmatrix} 0 & 0 \\ 4 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix} \quad (d) \begin{bmatrix} 3 & -2 & 7 \\ 0 & 0 & 3 \\ 0 & 0 & 8 \end{bmatrix}$$

3. Find the product by inspection (using the diagonal multiplication rule).

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -4 & 1 \\ 2 & 5 \end{bmatrix}$$

7. Find A^2 , A^{-2} , and A^{-k} (where k is any integer) by inspection.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$

8. Find A^2 , A^{-2} , and A^{-k} by inspection.

$$A = \begin{bmatrix} -6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

11. Compute the product by inspection.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

13. Compute the indicated power by inspection.

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}^{39}$$

15. Use the diagonal multiplication rule to compute the product by inspection.

$$\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} u & v \\ w & x \\ y & z \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} r & s & t \\ u & v & w \\ x & y & z \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

17. Create a symmetric matrix by substituting appropriate numbers for the $\times$'s.

$$(a) \begin{bmatrix} 2 & -1 \\ \times & 3 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & \times & \times & \times \\ 3 & 1 & \times & \times \\ 7 & -8 & 0 & \times \\ 2 & -3 & 9 & 0 \end{bmatrix}$$

19. Determine whether the matrix is invertible by inspection.

$$\begin{bmatrix} 0 & 6 & -1 \\ 0 & 7 & -4 \\ 0 & 0 & -2 \end{bmatrix}$$

21. Determine whether the matrix is invertible by inspection.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -5 & 0 & 0 \\ 4 & -3 & 4 & 0 \\ 1 & -2 & 1 & 3 \end{bmatrix}$$

23. Find the diagonal entries of AB by inspection.

$$A = \begin{bmatrix} 3 & 2 & 6 \\ 0 & 1 & -2 \\ 0 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & 7 \\ 0 & 5 & 3 \\ 0 & 0 & 6 \end{bmatrix}$$

25. Find all values of a (if any) for which A is symmetric.

$$A = \begin{bmatrix} 4 & -3 \\ a+5 & -1 \end{bmatrix}$$

26. Find all values of a, b, c for which A is symmetric.

$$A = \begin{bmatrix} 2 & a-2b+2c & 2a+b+c \\ 3 & 5 & a+c \\ 0 & -2 & 7 \end{bmatrix}$$

27. Find all values of x for which A is invertible.

$$A = \begin{bmatrix} x-1 & x^2 & x^4 \\ 0 & x+2 & x^3 \\ 0 & 0 & x-4 \end{bmatrix}$$

29. If A is an invertible upper triangular or lower triangular matrix, what can you say about the diagonal entries of A^{-1} ?

30. Show that if A is a symmetric $n \times n$ matrix and B is any $n \times m$ matrix, then the following products are symmetric: $B^T B$, BB^T , $B^T AB$.

31. Find a diagonal matrix A that satisfies $A^5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

33. Verify Theorem 1.7.1(b) for the matrix product AB and Theorem 1.7.1(d) for the matrix A , where

$$A = \begin{bmatrix} -1 & 2 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -8 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

34. Let A be an $n \times n$ symmetric matrix.

(a) Show that A^2 is symmetric.

(b) Show that $2A^2 - 3A + I$ is symmetric.

37. Let $A = [a_{ij}]$ be an $n \times n$ matrix. Determine whether A is symmetric.

(a) $a_{ij} = i^2 + j^2$ (b) $a_{ij} = i^2 - j^2$ (c) $a_{ij} = 2i + 2j$ (d) $a_{ij} = 2i^2 + 2j^3$

Exercise Set 1.8

1. Find the domain and codomain of $T_A(\mathbf{x}) = A\mathbf{x}$ when:

(a) A has size 3×2 . (b) A has size 2×3 . (c) A has size 3×3 . (d) A has size 1×6 .

3. Find the domain and codomain of the transformation defined by the equations.

(a) $w_1 = 4x_1 + 5x_2$, $w_2 = x_1 - 8x_2$

(b) $w_1 = 5x_1 - 7x_2$, $w_2 = 6x_1 + x_2$, $w_3 = 2x_1 + 3x_2$

5. Find the domain and codomain of T defined by the matrix product.

$$(a) \begin{bmatrix} 3 & 1 & 2 \\ 6 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad (b) \begin{bmatrix} 2 & -1 \\ 4 & 3 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

7. Find the domain and codomain of T defined by the formula.

(a) $T(x_1, x_2) = (2x_1 - x_2, x_1 + x_2)$ (b) $T(x_1, x_2, x_3) = (4x_1 + x_2, x_1 + x_2)$

9. Find the domain and codomain.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 4x_1 \\ x_1 - x_2 \\ 3x_2 \end{bmatrix}$$

11. Find the standard matrix for the transformation defined by the equations.

(a) $w_1 = 2x_1 - 3x_2 + x_3$, $w_2 = 3x_1 + 5x_2 - x_3$

(b) $w_1 = 7x_1 + 2x_2 - 8x_3$, $w_2 = -x_2 + 5x_3$, $w_3 = 4x_1 + 7x_2 - x_3$

13. Find the standard matrix for T defined by the formula.

(a) $T(x_1, x_2) = (x_2, -x_1, x_1 + 3x_2, x_1 - x_2)$

(b) $T(x_1, x_2, x_3, x_4) = (7x_1 + 2x_2 - x_3 + x_4, x_2 + x_3, -x_1)$

14. Find the standard matrix for the operator T defined by the formula.

(a) $T(x_1, x_2) = (2x_1 - x_2, x_1 + x_2)$ (d) $T(x_1, x_2, x_3) = (4x_1, 7x_2, -8x_3)$

15. Find the standard matrix for the operator $T : R^3 \rightarrow R^3$ defined by

$$w_1 = 3x_1 + 5x_2 - x_3, \quad w_2 = 4x_1 - x_2 + x_3, \quad w_3 = 3x_1 + 2x_2 - x_3$$

and compute $T(-1, 2, 4)$ by direct substitution and by matrix multiplication.

17. Find the standard matrix for T and use it to compute $T(\mathbf{x})$.

(a) $T(x_1, x_2) = (-x_1 + x_2, x_2)$; $\mathbf{x} = (-1, 4)$

(b) $T(x_1, x_2, x_3) = (2x_1 - x_2 + x_3, x_2 + x_3, 0)$; $\mathbf{x} = (2, 1, -3)$

19. Find $T_A(\mathbf{x})$ and express your answer in matrix form.

(a) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $\mathbf{x} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ (b) $A = \begin{bmatrix} -1 & 2 & 0 \\ 3 & 1 & 5 \end{bmatrix}$; $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$

21. Use Theorem 1.8.2 to show that T is a matrix transformation.

(a) $T(x, y) = (2x + y, x - y)$ (b) $T(x_1, x_2, x_3) = (x_1 + x_3, x_1 + x_2)$

23. Use Theorem 1.8.2 to show that T is *not* a matrix transformation.

(a) $T(x, y) = (x^2, y)$ (b) $T(x, y, z) = (x, y, xz)$

25. A function of the form $f(x) = mx + b$ is commonly called a “linear function” because its graph is a line. Is f a matrix transformation on R ? Explain.

27. The images of the standard basis vectors for R^3 under a linear transformation $T : R^3 \rightarrow R^3$ are given. Find the standard matrix for T and compute $T(\mathbf{x})$.

$$T(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad T(\mathbf{e}_3) = \begin{bmatrix} 4 \\ -3 \\ -1 \end{bmatrix}; \quad \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

29. Use matrix multiplication to find the reflection of $(-1, 2)$ about the

(a) x -axis. (b) y -axis. (c) line $y = x$.

31. Use matrix multiplication to find the reflection of $(2, -5, 3)$ about the

(a) xy -plane. (b) xz -plane. (c) yz -plane.

33. Use matrix multiplication to find the orthogonal projection of $(2, -5)$ onto the

(a) x -axis. (b) y -axis.

37. Use matrix multiplication to find the image of $(3, -4)$ when rotated about the origin through angle

(a) $\theta = 30^\circ$. (b) $\theta = -60^\circ$. (c) $\theta = 45^\circ$.

39. Let $T : R^2 \rightarrow R^2$ be a linear operator for which the images of the standard basis vectors are $T(\mathbf{e}_1) = (a, b)$ and $T(\mathbf{e}_2) = (c, d)$. Find $T(1, 1)$.

44. If multiplication by A rotates a vector $\mathbf{x}$ in the xy -plane through angle θ , what is the effect of multiplying $\mathbf{x}$ by A^T ? Explain your reasoning.

45. Find the standard matrix A for the linear transformation $T : R^2 \rightarrow R^2$ for which

$$T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ -2 \end{bmatrix}, \quad T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

50. (a) Prove: If $T : R^n \rightarrow R^m$ is a matrix transformation, then $T(\mathbf{0}) = \mathbf{0}$.

(b) Find an example of a mapping $T : R^n \rightarrow R^m$ for which $T(\mathbf{0}) = \mathbf{0}$ but T is not a matrix transformation.

Exercise Set 1.9

1. Determine whether the operators T_1 and T_2 commute (i.e., whether $T_1 \circ T_2 = T_2 \circ T_1$).
 - (a) $T_1 : R^2 \rightarrow R^2$ is the reflection about the line $y = x$, and $T_2 : R^2 \rightarrow R^2$ is the orthogonal projection onto the x -axis.
 - (b) $T_1 : R^2 \rightarrow R^2$ is the reflection about the x -axis, and $T_2 : R^2 \rightarrow R^2$ is the reflection about the line $y = x$.
2. Determine whether the operators commute.
 - (a) $T_1 : R^2 \rightarrow R^2$ is the orthogonal projection onto the x -axis, and $T_2 : R^2 \rightarrow R^2$ is the orthogonal projection onto the y -axis.
 - (b) $T_1 : R^2 \rightarrow R^2$ is the rotation through angle $\pi/4$, and $T_2 : R^2 \rightarrow R^2$ is the reflection about the y -axis.
3. $T_1 : R^3 \rightarrow R^3$ is the reflection about the xy -plane, and $T_2 : R^3 \rightarrow R^3$ is the orthogonal projection onto the yz -plane. Determine whether T_1 and T_2 commute.
4. $T_1 : R^3 \rightarrow R^3$ is the reflection about the xy -plane, and $T_2 : R^3 \rightarrow R^3$ is defined by $T_2(x, y, z) = (2x, 3y, z)$. Determine whether T_1 and T_2 commute.
5. Let T_A and T_B be operators whose standard matrices are given. Find the standard matrices for $T_B \circ T_A$ and $T_A \circ T_B$.
$$A = \begin{bmatrix} 1 & -2 \\ 4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -3 \\ 5 & 0 \end{bmatrix}$$
7. Find the standard matrix for the stated composition in R^2 .
 - (a) A rotation of 90° , followed by a reflection about the line $y = x$.
 - (b) An orthogonal projection onto the y -axis, followed by a 45° counterclockwise rotation.
 - (c) A reflection about the x -axis, followed by a rotation of 60° .
8. Find the standard matrix for the stated composition in R^2 .
 - (a) A rotation of 60° , followed by an orthogonal projection onto the x -axis, followed by a reflection about the line $y = x$.
 - (b) A rotation of 15° , followed by a rotation of 105° , followed by a rotation of 60° .
9. Find the standard matrix for the stated composition in R^3 .
 - (a) A reflection about the yz -plane, followed by an orthogonal projection onto the xz -plane.
 - (b) A reflection about the xy -plane, followed by an orthogonal projection onto the xy -plane.
11. Let $T_1(x_1, x_2) = (x_1 + x_2, x_1 - x_2)$ and $T_2(x_1, x_2) = (3x_1, 2x_1 + 4x_2)$.
 - (a) Find the standard matrices for T_1 and T_2 .
 - (b) Find the standard matrices for $T_2 \circ T_1$ and $T_1 \circ T_2$.
 - (c) Use the matrices from (b) to find formulas for $T_1(T_2(x_1, x_2))$ and $T_2(T_1(x_1, x_2))$.
13. Let $T_1(x_1, x_2) = (x_1 - x_2, 2x_2 - x_1, 3x_1)$ and $T_2(x_1, x_2, x_3) = (4x_2, x_1 + 2x_2)$.
 - (a) Find the standard matrices for T_1 and T_2 .
 - (b) Find the standard matrix for $T_2 \circ T_1$ and $T_1 \circ T_2$.
 - (c) Use the matrices from (b) to find formulas for $T_1(T_2(x_1, x_2, x_3))$ and $T_2(T_1(x_1, x_2))$.

17. Express the equations in matrix form; then use Theorem 1.5.3(c) to determine whether the operator defined by the equations is invertible.
- (a) $w_1 = 8x_1 + 4x_2$, $w_2 = 2x_1 + x_2$
- (b) $w_1 = -x_1 + 3x_2 + 2x_3$, $w_2 = 2x_1 + 4x_3$, $w_3 = x_1 + 3x_2 + 6x_3$
19. Determine whether the operator $T : R^2 \rightarrow R^2$ is invertible; if so, find the standard matrix for T^{-1} and find $T^{-1}(w_1, w_2)$.
- (a) $w_1 = x_1 + 2x_2$, $w_2 = -x_1 + x_2$
- (b) $w_1 = 4x_1 - 6x_2$, $w_2 = -2x_1 + 3x_2$
21. Determine whether the matrix operator is invertible. If so, describe the effect of its inverse in words.
- (a) Reflection about the x -axis in R^2 .
- (b) A 60° counterclockwise rotation about the origin in R^2 .
- (c) Orthogonal projection onto the x -axis in R^2 .
23. Determine whether the operator T_A is invertible. If so, compute $T_A^{-1}(\mathbf{x})$.
- (a) $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$; $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- (b) $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$; $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$
25. Let $T_A : R^2 \rightarrow R^2$ be multiplication by
- $$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$
- (a) Describe the geometric effect of applying T_A to a vector $\mathbf{x}$ in R^2 .
- (b) Express the operator T_A as a composition of two linear operators on R^2 .
27. Prove that the matrix transformations T_A and T_B commute if and only if the matrices A and B commute.
28. Let T_A and T_B be matrix operators on R^n . Prove that $T_A \circ T_B$ is invertible if and only if both T_A and T_B are invertible.
29. Prove that the matrix operator T_A on R^n is invertible if and only if for every vector $\mathbf{b}$ in R^n there exists a unique vector $\mathbf{x}$ in R^n such that $T_A(\mathbf{x}) = \mathbf{b}$.

Exercise Set 2.1

1. Find all minors and cofactors of $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 7 & -1 \\ -3 & 1 & 4 \end{bmatrix}$.
3. Let $A = \begin{bmatrix} 4 & -1 & 1 & 6 \\ 0 & 0 & -3 & 3 \\ 4 & 1 & 0 & 14 \\ 4 & 1 & 3 & 2 \end{bmatrix}$. Find: (a) M_{13} and C_{13} (b) M_{23} and C_{23} (c) M_{22} and C_{22} (d) M_{21} and C_{21}

5. Evaluate $\det(A) = \begin{vmatrix} 3 & 5 \\ -2 & 4 \end{vmatrix}$. If A is invertible, find A^{-1} using Formula (2).

7. Evaluate $\det(A) = \begin{vmatrix} -5 & 7 \\ -7 & -2 \end{vmatrix}$. If A is invertible, find A^{-1} .

In Exercises 9–11, use the arrow technique to evaluate the determinant.

9. $\begin{vmatrix} a-3 & 5 \\ -3 & a-2 \end{vmatrix}$

11. $\begin{vmatrix} -2 & 1 & 4 \\ 3 & 5 & -7 \\ 1 & 6 & 2 \end{vmatrix}$

In Exercises 15–17, find all values of λ for which $\det(A) = 0$.

15. $A = \begin{bmatrix} \lambda-2 & 1 \\ -5 & \lambda+4 \end{bmatrix}$

17. $A = \begin{bmatrix} \lambda-1 & 0 \\ 2 & \lambda+1 \end{bmatrix}$

19. Evaluate the determinant of the matrix in Exercise 13, $A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{bmatrix}$, by cofactor expansion

along: (a) the first row, (b) the first column, (c) the second row, (d) the second column, (e) the third row, (f) the third column.

In Exercises 21–25, evaluate $\det(A)$ by cofactor expansion along a row or column of your choice.

21. $A = \begin{bmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{bmatrix}$

24. $A = \begin{bmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{bmatrix}$

25. $A = \begin{bmatrix} 3 & 3 & 0 & 5 \\ 2 & 2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$

In Exercises 27–32, evaluate $\det(A)$ by inspection.

27. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

30. $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

32. $A = \begin{bmatrix} -3 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 40 & 10 & -1 & 0 \\ 100 & 200 & -23 & 3 \end{bmatrix}$

33. In each part, show that the value of the determinant is independent of θ .

(a) $\begin{vmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{vmatrix}$

(b) $\begin{vmatrix} \sin \theta & \cos \theta & 0 \\ -\cos \theta & \sin \theta & 0 \\ \sin \theta - \cos \theta & \sin \theta + \cos \theta & 1 \end{vmatrix}$

36. Show that $\det(A) = \frac{1}{2} \begin{vmatrix} \operatorname{tr}(A) & 1 \\ \operatorname{tr}(A^2) & \operatorname{tr}(A) \end{vmatrix}$ for every 2×2 matrix A .

37. What can you say about an n th-order determinant all of whose entries are 1? Explain.

38. What is the maximum number of zeros that a 3×3 matrix can have without having a zero determinant? Explain.

Working with Proofs

40. Prove that (x_1, y_1) , (x_2, y_2) , (x_3, y_3) are collinear if and only if $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$.

43. A matrix of the form $V = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$ (and its higher-order analogs) is called a **Vandermonde matrix**. Use cofactor expansion to prove that

$$\begin{vmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{vmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$$

Exercise Set 2.2

In Exercises 1–4, verify that $\det(A) = \det(A^T)$.

1. $A = \begin{bmatrix} -2 & 3 \\ 1 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 2 & 4 \\ 5 & -3 & 6 \end{bmatrix}$

In Exercises 5–8, find the determinant of the given elementary matrix by inspection.

5. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$7. \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In Exercises 9–13, evaluate the determinant of the matrix by first reducing to row echelon form and then using some combination of row operations and cofactor expansion.

$$9. \begin{bmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{bmatrix}$$

$$11. \begin{bmatrix} 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

$$13. \begin{bmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

In Exercises 15–22, evaluate the determinant, given that $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$.

$$15. \begin{vmatrix} d & e & f \\ g & h & i \\ a & b & c \end{vmatrix}$$

$$17. \begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix}$$

$$20. \begin{vmatrix} a & b & c \\ 2d & 2e & 2f \\ g+3a & h+3b & i+3c \end{vmatrix}$$

$$22. \begin{vmatrix} a & b & c \\ d & e & f \\ 2a & 2b & 2c \end{vmatrix}$$

23. Use row reduction to show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

24. Verify the formulas in parts (a) and (b) and then conjecture a general result.

$$(a) \det \begin{bmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = -a_{13}a_{22}a_{31}$$

$$(b) \det \begin{bmatrix} 0 & 0 & 0 & a_{14} \\ 0 & 0 & a_{23} & a_{24} \\ 0 & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} = a_{14}a_{23}a_{32}a_{41}$$

In Exercises 25 and 27, confirm the identity without evaluating any of the determinants directly.

$$25. \begin{vmatrix} a_1 + b_1 & b_1 + c_1 & c_1 \\ a_2 + b_2 & b_2 + c_2 & c_2 \\ a_3 + b_3 & b_3 + c_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$27. \begin{vmatrix} a_1 + b_1 & a_1 - b_1 & c_1 \\ a_2 + b_2 & a_2 - b_2 & c_2 \\ a_3 + b_3 & a_3 - b_3 & c_3 \end{vmatrix} = -2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

In Exercises 29–30, show that $\det(A) = 0$ without directly evaluating the determinant.

$$29. A = \begin{bmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & 3 \end{bmatrix}$$

33. Let A be an $n \times n$ matrix, and let B be the matrix that results when the rows of A are written in reverse order. State a theorem that describes how $\det(A)$ and $\det(B)$ are related.

34. Find the determinant of the matrix

$$\begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}$$

Exercise Set 2.3

In Exercises 1–3, verify that $\det(kA) = k^n \det(A)$.

$$1. A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}; \quad k = 2$$

$$3. A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}; \quad k = -2$$

In Exercise 5, verify that $\det(AB) = \det(BA)$ and determine whether $\det(A+B) = \det(A) + \det(B)$.

$$5. A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 3 \\ 7 & 1 & 2 \\ 5 & 0 & 1 \end{bmatrix}$$

In Exercises 7–14, use determinants to decide whether the matrix is invertible.

$$7. A = \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$$

$$9. A = \begin{bmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{bmatrix}$$

$$11. A = \begin{bmatrix} 4 & 2 & 8 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{bmatrix}$$

$$14. A = \begin{bmatrix} \sqrt{2} & -\sqrt{7} & 0 \\ 3\sqrt{2} & -3\sqrt{7} & 0 \\ 5 & -9 & 0 \end{bmatrix}$$

In Exercises 15–17, find the values of k for which the matrix A is invertible.

$$15. A = \begin{bmatrix} k-3 & -2 \\ -2 & k-2 \end{bmatrix}$$

$$17. A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 6 \\ k & 3 & 2 \end{bmatrix}$$

In Exercises 19 and 22, decide whether the matrix is invertible; if so, use the adjoint method to find its inverse.

$$19. A = \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$$

$$22. A = \begin{bmatrix} 2 & 0 & 0 \\ 8 & 1 & 0 \\ -5 & 3 & 6 \end{bmatrix}$$

In Exercises 24–27, solve by Cramer's rule, where it applies.

$$24. 7x_1 - 2x_2 = 3, \quad 3x_1 + x_2 = 5$$

$$25. 4x + 5y = 2, \quad 11x + y + 2z = 3, \quad x + 5y + 2z = 1$$

$$27. x_1 - 3x_2 + x_3 = 4, \quad 2x_1 - x_2 = -2, \quad 4x_1 - 3x_3 = 0$$

$$30. \text{ Show that } A = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is invertible for all values of } \theta; \text{ then find } A^{-1} \text{ using Theo-}$$

rem 2.3.6.

$$33. \text{ Let } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \text{ with } \det(A) = -7. \text{ Find:}$$

$$(a) \det(3A)$$

(b) $\det(A^{-1})$

(c) $\det(2A^{-1})$

(d) $\det((2A)^{-1})$

(e) $\det \begin{bmatrix} a & g & d \\ b & h & e \\ c & i & f \end{bmatrix}$

34. In each part, find the determinant given that A is a 4×4 matrix with $\det(A) = -2$:

(a) $\det(-A)$

(b) $\det(A^{-1})$

(c) $\det(2A^T)$

(d) $\det(A^3)$